

Question	Marking Guidance	Mark	Comments
04.1	1. Change in sequence of <u>DNA</u> bases OR <u>DNA</u> base sequence inverted/reversed; 2. Change in (sequence of) amino acid(s) OR Change in primary structure; 3. Change in hydrogen/ionic/disulfide bonds; 4. Alters tertiary/3° structure (of protein);	4 (4 x AO2)	1. Accept if answer refers to DNA and shows base sequence inverted e.g. TATGC to CGTAT 1. Accept nucleotides for bases 1. Accept changes DNA triplet(s) 2. Reject amino acids are 'formed' or 'produced' 2. Accept shorter amino acid sequence or terminated 4. Reject active site/enzyme 4. Ignore quaternary 4. Ignore 3D

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04.2	Mark point 1 required for maximum marks. Max 3 marks from mark points 4 to 7 (For) 1. SDs do not overlap so significant difference/decrease/change (with Roctavian); 2. Only one injection/treatment is required OR Weekly/monthly/regular treatments/injections not required; 3. (Investigation) compares same volunteers; (Against) 4. Only males (investigated/treated) OR No females (investigated/treated); 5. (Possible) side effects; 6. At least 18 (years of age investigated/treated) OR No under 18 (years of age investigated/treated); 7. Only severe HA (investigated/treated);	5 max (5 x AO3)	1. Reject 'results are significant' 1. Accept 'not due to chance' for 'significant' 2. Accept fewer/less for 'one' Ignore 'sample size' 'length of investigation' and 'repeat studies required' Ignore reference to statistical test 6. Accept children/young for under 18 6. Reject only 18 years of age treated

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04.3	0.125 / 12.5% / $\frac{1}{8}$;	1 (1 x AO2)	Reject 1 : 8 or 8 : 1 Accept 1 in 8 Reject 12.5 without %
